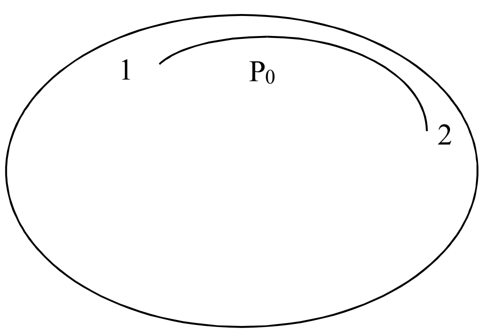
GEOMATICS
NTNU

Version 2010

TBA4245 GEODESY

Example exercise no. 3: The Geodesic. The two geodetic main tasks. Here: The 2nd geodetic main task

An airline company is planning to establish a regular air service between Gardermoen airport Oslo and Tokyo, and the company wants to find the shortest distance between these two places. The motivation is to save time and fuel.
In this exercise we disregard the height and assume the aeroplane to be moving along the ellipsoidal surface.



Here the following very approximate values of geographical latitude and longitude are used:

Point 1: Garderm.: $B_1 = 60°$ N $L_1 = 11°$ E Gr.w.
Point 2: Tokyo: $B_2 = 35°$ N $L_2 = 140°$ E Gr.w.

(a) Consider which datum you will use. Substantiate your choice!
Calculate manually the length of the route?

(b) Calculate manually which course (azimuth) the aeroplane must have at the starting point of the flight in Gardermoen.

(c) Calculate manually which course (azimuth) the aeroplane must have when it is returning from Tokyo.

(d) Calculate manually the latitude of the northernmost point of the geodesic going through Gardermoen and Tokyo. *Hint: Use one of Clairiaut's equations in chapter 1 in Geodesy part 2.*

(e) Calculate manually the length of the arc of the normal section from Gardermoen to Tokyo and from Tokyo to Gardermoen. Why will we generally obtain two different values of these two distances?

(f) Calculate manually the difference in the points Gardermoen and Tokyo between the azimuth of the geodesic and the azimuth of the arc of the normal section. Comment the results in (e) and (f).

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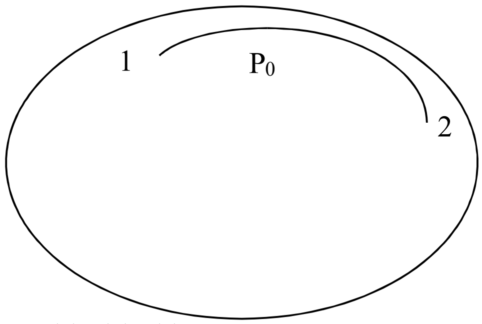
Example exercise no (3): Solution:



Geographical (ellipsoidal) latitude and longitude:

Point 1: Garderm.:
$B_1 = 60°$ N $L_1 = 11°$ E Gr.w.
Point 2: Tokyo:
$B_2 = 35°$ N $L_2 = 140°$ E Gr.w.

(a), **(b)**, **(c)**:

The 2nd Geodetic main task gives the answers of the questions in (a), (b) and (c).

The formulas: See chapter 2.6
Another example is given in the last pages of chap. 2 in Holsen Geodesy part 2.

The Holsen program L-GEO2 uses the formulas in 2.6 for exact computations. *See a printout of using the program L-GEO2.*

The datum chosen is the global datum WGS84. The spheroidal semi axes of the spheroid defined in WGS84.

=> Ellipsoidal parameters: $a = 6378137{,}0000$ m
$b = 6356752{,}3142$
$e^2 = 0{,}006694380$

Assuming the aeroplane to be moving along the spheroid surface

(a) The length of the route?
The 2nd geodetic main task, chapter 2.6.
The geodesic gives the shortest distance.

Reduced latitudes: $\beta_1 = 59.91660780°$
$\beta_2 = 34.90964204°$

Use Matlab, spread sheet (Excel?) or similar, as we have to do several iterations before getting the final answer.

The calculations including the intermediate answers are shown in 2 tables.
3 iterations gave the same answers as the Holsen program L-GEO2.

The series of the formulas are developed in point P_0 on the sphere. (in P_0 the azimuth is 90°.)

For the final values of β_0, σ_1, σ_2, λ_1 and λ_2, see the tables.

Table 1:

	Starting values	**Column 3:** **1st iteration**	**2nd iteration**	**3rd iteration**
ΔL	129.000000000°	129.000000000°	129.000000000°	129.000000000°
(i) $= R_1 \cdot \Delta\sigma^{\text{radians}}$		1.3331632605	1.3337464011	1.3337474693
(ii) $= R_2 \cdot (\quad - \quad)$		$6.369481 \cdot 10^{-4}$	$6.364579 \cdot 10^{-4}$	$6.364587 \cdot 10^{-4}$
(iii) $= R_3 \cdot (\quad - \quad)$		$6.7947 \cdot 10^{-8}$	$6.7999 \cdot 10^{-8}$	$6.7999 \cdot 10^{-8}$
(iv) = (i)-(ii)+(iii) $\Delta\sigma$ in radians		1.3325263804	1.3331100112	1.3331110786
(v) = (iv) $\cdot$ *R* i grader		0.083761853°	0.083948192°	0.083948016°
$\Delta\lambda = \Delta L + (v)$	$\Delta\lambda = \Delta L \cdot V =$ 129.19818545°	129.083761853°	129.08394819°	129.08394802°
Ergo:				
λ_1	-53.20937178	-53.12356535	-53.12370510	-53.12370497°
λ_2	+75.98881367	+75.96019650	+75.96024309	+75.9602430°
β_0	70.86660014	70.83109738	70.83115516	70.83115511°
σ_1	-23.66720917	-23.63905938	-23.63910527	-23.63910523°
σ_2	52.71741906	52.70803088	52.70804617	52.70804616°
$\Delta\sigma$	76.38462823	76.34709026	76.34715145	76.34715139°
	Go to 2nd column in table 2	Go to 3rd column in table 2	Go to 4th column in table 2	Go to 5th column in table 2

Table 2:

	Column 2: From the start values in column 2 in table 1	**From the 1st iteration values**	**From the 2nd iteration values**	**From the 3rd iteration values**
R	$1.097104080 \cdot 10^{-3}$	$1.0990633472 \cdot 10^{-3}$	$1.0990601589 \cdot 10^{-3}$	
B_0	70.92610823°	70.890698996°	70.89075662°	70.89075657°
W_0	0.99700577347	0.997007056064	0.9970070540	0.997007053977
k_1	$1.4993579747 \cdot 10^{-3}$	$1.4987147526 \cdot 10^{-3}$	$1.498715800 \cdot 10^{-3}$	$1.49871579924 \cdot 10^{-3}$
n	$1.6792203899 \cdot 10^{-3}$	$1.6792203899 \cdot 10^{-3}$	$1.6792203899 \cdot 10^{-3}$	
R_1	1.00092897938	1.00092930148	1.00092930095	
R_2	$3.7483949 \cdot 10^{-4}$	$3.7467869 \cdot 10^{-4}$	$3.7467895 \cdot 10^{-4}$	
R_3	$1.405046 \cdot 10^{-7}$	$1.40384 \cdot 10^{-7}$	$1.40384 \cdot 10^{-7}$	
	Go to 3rd column in table 1 (1st iteration)	Go to 4th column in table 1 (2nd iteration)	Go to 5th column in table 1 (3rd iteration)	

When the final values from the iterations, we can calculate the azimuth and the length of the geodesic:

Azimuth:

Azimuth from Gardermoen to Tokyo: $A_{1\text{-}2}$ $A_1 = \text{inv}\tan\left(\dfrac{-1}{\sin\sigma_1 \cdot \tan\beta_0}\right) = 40.92385877°$

Azimuth from Tokyo to Gardermoen of the geodesic. We choose the "correct" solution of the 2 invtan solutions:

$$A_2 = \text{inv}\tan\left(\frac{-1}{\sin\sigma_2 \cdot \tan\beta_0}\right) = 336.39659740°$$

The length of the geodesic:

For values of the variables W_0 and k_1 , see table 2, the last column. We obtain:

$c = 6366297.1539$
$D_1 = 7.4935727 \cdot 10^{-4}$
$D_2 = 1.403843 \cdot 10^{-7}$
$D_3 = 7.01321 \cdot 10^{-11}$

The distance from P_0 to Gardermoen (= point 1), a negative value because Gardermoen is west of P_0: $S_1 = -2630112.1558$ m

The distance from P_0 to Tokyo (= point 2): $S_2 = 5861140.4871$ m

The length of the route:
$\Delta S = +2630112.1558 + 5861140.4871 = 8491\ 252.643$m

(d) *The northernmost point?*
B_0 is computed in (a) to be 70.89075657°.

We can check the result by using …?

(e) *Difference in azimuth and distance between the geodesic(G) and the arc of the normal section (NS)?*

Azimuth (formula 1.6):
In Gardermoen: $diff = A_G - A_{NS} = -49.96093'' = -0.013878°$
In Tokyo: $diff = A_G - A_{NS} = +99.40645'' = +0.027613°$

Distance(formula 1.7):
$e'^2 = 0.00673949676$ (for WGS84 ellipsoid). The geodesic is always the shortest line:
From Gardermoen to Tokyo:
$\eta^2 = 0.00168487419$ $N = 6394209.17$
which gives: $diff = 0.204$ m

From Tokyo to Gardermoen:
$\eta^2 = 0.0045222702016$ $N = 6385172.17$

which gives: *diff* = 0.812 m

The trace of the normal sections are different in the two end points, see chapter 1.

Holsens L-GEO2: The 2nd geodetic main task

Before computing you must answer with an integer your choice of datum, and for some programs your choice of which way to compute and what type of coordinates (UTM or latitude/longitude).

Another way to run the program: Create shortcut (right hand click and then choose).

If you have problems using the Holsen's programs, try this:

Find and choose the actual program.

NB: Right hand click, choose Properties and check if the box "Close at exit" is turned off.